

Just Noticeable Difference for Images with Decomposition Model for Separating Edge and Textured Regions

Anmin Liu, Weisi Lin, Manoranjan Paul, Chenwei Deng, and Fan Zhang

Abstract—In just noticeable difference (JND) models, evaluation of contrast masking (CM) is a crucial step. More specifically, CM due to edge masking (EM) and texture masking (TM) needs to be distinguished due to the entropy masking property of the human visual system. However, TM is not estimated accurately in the existing JND models since they fail to distinguish TM from EM. In this letter, we propose an enhanced pixel domain JND model with a new algorithm for CM estimation. In our model, total-variation based image decomposition is used to decompose an image into structural image (i.e., cartoon like, piecewise smooth regions with sharp edges) and textural image for estimation of EM and TM, respectively. Compared with the existing models, the proposed one shows its advantages brought by the better EM and TM estimation. It has been also applied to noise shaping and visual distortion gauge, and favorable results are demonstrated by experiments on different images.

Index Terms—Contrast masking, entropy masking, just noticeable difference (JND), total variation (TV), visual distortion gauge.

I. INTRODUCTION

As we know, the human visual system (HVS) is only capable of perceiving the pixel change above a certain visibility threshold determined by the underlying physiological and psychophysical mechanisms. Therefore, we should consider not only statistical properties among pixels but also the perceptual features in image processing since the HVS is the ultimate receiver of the majority of processed images. Just noticeable difference (JND), which accounts for the maximum sensory distortion that the typical HVS does not perceive (e.g., for 75% of the observers), can be adopted in perceptual image/video processing algorithms and systems.

The JND is of the same size with the image itself and its values can be estimated for the pixel or subband (e.g., discrete cosine transform) domain. Two images cannot be visually distinguished if the difference in each image pixel (for pixel

domain JND) or image subband (for subband JND) is within the JND value. In image/video compression, JND can be used to improve bit allocation [1], [2] to remove the perceptual redundancy. In watermarking [3], visual quality evaluation [4], and image enhancement [5], JND were also shown to be effective.

In the applications such as visual quality evaluation and image enhancement, a JND estimator directly from the pixel domain is more convenient, because the transformation from subbands is avoided; thereafter, in this letter, we focus on JND estimators in pixel domain. This category of JND models basically take into account two factors: luminance adaptation (LA, which refers to the masking effect of the HVS toward background luminance) and contrast masking (CM, which denotes the visibility reduction of one visual signal at the presence of another one [6]). As an extension of Chou and Li's model in [13], Yang *et al.*'s model [1] is the latest direct pixel domain JND model, which accounts for the difference between texture masking (TM, i.e., CM due to texture) and edge masking (EM, i.e., CM due to edge).

Edge structure is simpler and more predictable than a texture one, and a typical observer has prior knowledge (i.e., related to a specific object) about what edges should look like [7], [15]. Therefore, distinguishing between TM and EM is very important for JND estimation. One experiment in [7] has shown that three times as much quantization noise can be masked/hidden in the texture image than in the edge image without compromising perceptual quality even if both images have a similar power spectrum, background luminance level, and energy content in the local regions.

TM is usually underestimated (e.g., in Yang *et al.*'s model) since many textured regions tend to be classified as edge ones (to be detailed in Fig. 5). In this letter, we propose a new pixel domain JND model to solve the aforementioned drawback in the EM and TM estimations due to difficulty caused by no or not effective separation of edge and texture. In our model, total variation [8] (TV)-based image decomposition is used to decompose an image into structural image (being piecewise smooth and with sharp edge) for EM estimation and textural image for TM estimation, respectively; therefore, EM and TM can be effectively distinguished. With the experimental results, we confirm that the proposed model correlates better with the HVS perception with the evidence of an average additional peak signal-to-noise ratio (PSNR) redundancy of 1.13 dB from

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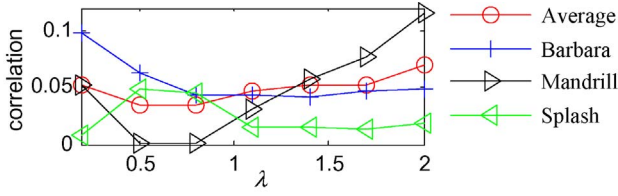


Fig. 1. Determination of λ for three images and the average result over the ten test images in Table I.

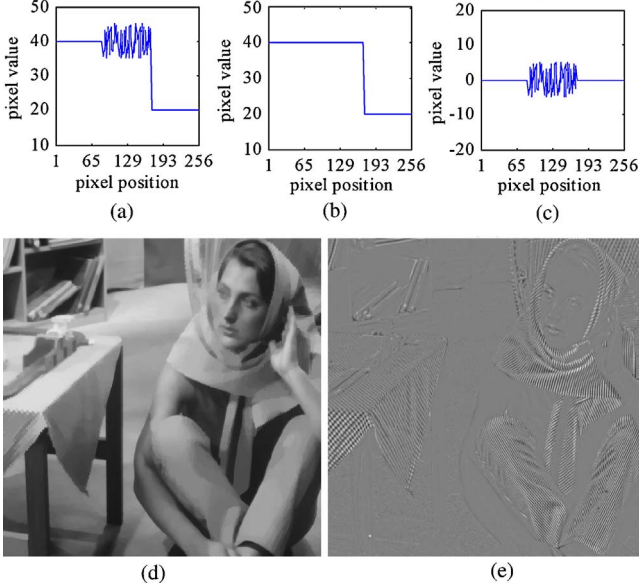


Fig. 2. Structure-texture decomposition results for TV-L¹ model. (a) 1-D signal. (b) Structural component of the original signal. (c) Textural component of the original signal. (d) Structural *Barbara* image [the original *Barbara* image is shown in Fig. 5(a)]. (e) Textural *Barbara* image (the value of 128 was added to the result for clear viewing).

the latest relevant model [1] and the improvement over visual quality gauging.

The rest of this letter is organized as follows. In Section II, we describe the problem of image decomposition modeling. Section III presents the details of the proposed pixel domain JND model. We discuss how to estimate TM and EM, and why the proposed method is better than the existing ones. Experimental results are given and discussed in Section IV while Section V concludes the letter.

II. IMAGE DECOMPOSITION MODEL

Image decomposition is to split an image into two or more component images. An image f can be regarded as the sum of the structural image u (being piecewise smooth and with sharp edge along the contour) and the textural image v (only containing fine-scale details, usually with some oscillatory nature), i.e., $f = u + v$. Decomposition is important for many image-processing applications, e.g., image coding, texture discrimination, image denoising, image inpainting, and image registration. A general approach for such decomposition is to solve the problem formulated in [8] as follows:

$$\min_u \{ \|s(u)\|_A \mid \|t(u, f)\|_B \leq \sigma \} \quad (1)$$

where $s(\cdot)$ and $t(\cdot, \cdot)$ are two functions to be chosen, and $\|\cdot\|_A$ and $\|\cdot\|_B$ are certain norms (or semi-norms). The fidelity term $\|t(u, f)\|_B \leq \sigma$ forces u to be close to f , and $t(u, f)$ is often chosen to be $(f - u)$ which is ideally the texture image v . The constrained minimization problem (1) is equivalent to its Lagrangian relaxed form “ $\min \{ \|s(u)\|_A + \lambda \|t(u, f)\|_B \}$ over u ” since $\|s(u)\|_A$ and $\|t(u, f)\|_B$ are convex in u , where λ is the Lagrange multiplier.

The TV-based method has been widely used in image structure-texture decomposition models. In such a model, $\|s(u)\|_A$ is chosen as $TV(u)$, where $TV(u) = \int |\nabla u|$ and ∇u denotes the generalized derivative (i.e., gradient) [8] of u . Minimizing $TV(u)$ allows u to have discontinuities; hence, edges in the original image are preserved [8]. In [9], different TV-based image decomposition models are considered and the model of minimizing TV with an L¹-norm fidelity term is shown to achieve better results; we adopt this (TV-L¹) model in our work for image decomposition, and then (1) becomes as follows:

$$\min_u \int_{\Omega} |\nabla u| + \lambda \int_{\Omega} |f - u| \quad (2)$$

where Ω is the domain of f .

In (2), the larger the λ is, the smaller the fidelity term becomes; this means that u is less smooth and closer to f . Aujol *et al.* [10] proposed the use of the correlation between u and v to estimate λ by finding a minimum of this correlation, and the typical value of λ is between 0.2 to 2 [8], [9] for most natural images. Fig. 1 shows the said correlation with λ values for three images. As can be seen, the optimal λ is different for different images, and it is computationally expensive to find the optimal value for each image. In this letter, a reasonable λ value 0.8 (see the average curve in Fig. 1) has been used for the sake of computational efficiency (for noncritical timing applications, the method in [10] can be used). In Fig. 2, we have shown the experimental results of decomposition for both a 1-D signal and a real-world image (*Barbara* image), with the TV-L¹ model. From Fig. 2(a)–(c) we can see that the structural component (the boundary of change) and the textural component (the small variation) can be well decomposed for a 1-D signal; from Fig. 2(d) and (e) we can see that the structural and the textural component images are reasonably decomposed for a real image.

III. PROPOSED JND MODEL

The proposed JND model is shown in Fig. 3. The parts enclosed with dashed lines represent the modules for CM which is our contribution in this letter. The other two modules, i.e., LA module and nonlinear additivity model for masking (NAMM) module (to be briefly described in Sections III-A and III-C below) are the same with those in [1]. Toward CM estimation, we evaluate EM and TM on u and v , respectively, rather than on the whole input image as in the existing models. Therefore, the problem mentioned in Section I for CM estimation in the existing direct pixel domain JND models can be avoided since edge and texture are distinguished by the TV-based decomposition.

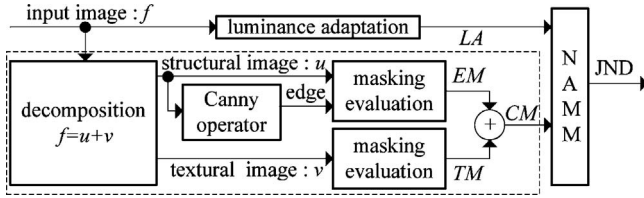


Fig. 3. Block diagram of the proposed direct pixel domain JND model.

A. Luminance Adaptation

LA refers to the masking effect of the HVS toward background luminance. The formula for LA is defined in [1] as follows:

$$LA(x, y) = \begin{cases} 17 \times (1 - \sqrt{f(x, y)/127}) + 3, & \text{if } f(x, y) \leq 127 \\ 3 \times (f(x, y) - 127)/128 + 3, & \text{otherwise} \end{cases} \quad (3)$$

where (x, y) represents the image pixel position.

B. Contrast Masking

CM is an important phenomenon in the HVS perception and is referred to the reduction in the visibility of one visual component at the presence of another [6]. Usually noise becomes less visible in the regions with high spatial variation, and more visible in smooth areas. CM in pixel domain includes TM and EM since both texture and edge exhibit high spatial variation; the value of CM on a smooth region is nearly zero since little spatial variation appears there.

Watson *et al.* [11] introduced the concept of entropy masking, which is due to unfamiliarity of the observer to visual content. Entropy masking is due to the limitation of the human brain in processing simultaneous complex phenomena [12]. As we know, a texture is less predictable and more difficult to learn than an edge [7]. Therefore, the effect of TM should be larger than that of EM if the corresponding regions have the same level of spatial variation.

In our model, CM can be represented as follows:

$$CM(x, y) = EM^u(x, y) + TM^v(x, y) \quad (4)$$

$$\begin{cases} EM^u(x, y) = C_s^u(x, y) \cdot \beta \cdot W_e \\ TM^v(x, y) = C_s^v(x, y) \cdot \beta \cdot W_t \end{cases} \quad (5)$$

where EM^u and TM^v are EM and TM in u and v , respectively, C_s^u and C_s^v are spatial contrasts for u and v , respectively, and C_s is the maximum luminance difference within the 5×5 neighborhood [13]. W_e and W_t are used to distinguish EM from TM, $W_e < W_t$ since TM results in a higher JND value than EM for a same extent of variation (C_s), and their values are set as 1 and 3 [7], respectively; the constant β is set as 0.117 as in [1].

Fig. 4 shows the detected edge information with Yang *et al.*'s model (i.e., via Canny detector [18]) and the proposed one (i.e., decomposition + Canny operator). Let Ω_{e-p}^f and Ω_{e-y}^f be the edge pixels in the proposed model and Yang *et al.*'s model. In the proposed model, $\Omega_{e-p}^f = \Omega_{e-y}^f$; in Yang *et al.*'s model, Canny operator with a threshold (denoted as t) is used on f . The larger the t is, the less the detected edge region

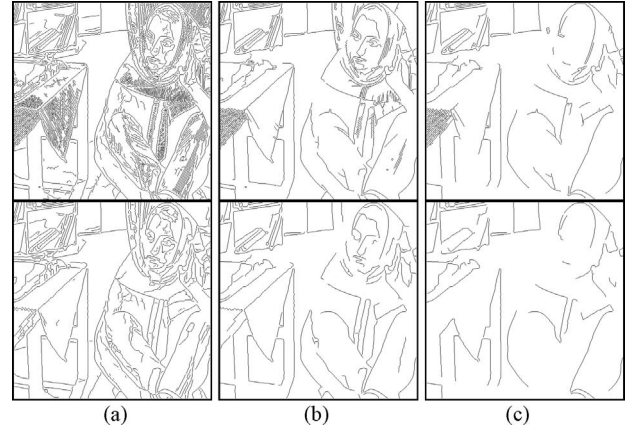


Fig. 4. Detected edge information (binary image, with black pixels representing edges) for the proposed and Yang *et al.*'s models, with different threshold (t), for *Barbara* image [as shown in Fig. 5(a)]; the first row is with Canny operator (as in Yang *et al.*'s model) and the second row is with the proposed decomposition and Canny operator. (a) $t = 0.1$. (b) $t = 0.3$. (c) $t = 0.5$.

becomes. For fair comparison, we use the same Canny operator (and the same t) on u in the proposed model.

From the figure, by using the proposed model, we distinguish between EM and TM better. We can see that true edges can be detected in both Yang *et al.*'s method and the proposed method. However, with Yang *et al.*'s method, some texture is detected alongside the edges with various values for t (e.g., the middle left part of the *Barbara* image in the top row of Fig. 4). Canny operator finds intensity variation over local neighborhoods [14], and is not designed for distinguishing edge and texture. However, by separating first the textural image from the image, we have higher chance to detect objects' true edges. Therefore, the proposed method (i.e., decomposition + Canny) outperforms the method in Yang *et al.*'s model (i.e., with only Canny operator).

To further illustrate the advantage of the proposed CM model, we have shown the CM profile for Yang *et al.*'s model and the proposed model in Fig. 5. Two typical images (*Barbara* and *Mandrill*) are used. From Fig. 5, we see that there is underestimation on TM in Yang *et al.*'s model (the second row in the figure), especially with the bottom-left of the desk cloth of *Barbara* image, and the cheek of *Mandrill* image, because the merely Canny-based method can hardly distinguish between edge and texture as we have discussed above; on the contrary, the proposed model produced reasonable CM values.

C. Overall JND Model

To integrate LA and CM for overall JND estimation, we adopt Yang's NAMM [1] (shown in Fig. 3), which can be described mathematically as follows:

$$JND = LA + CM - C^{lc} \times \min \{LA, CM\} \quad (6)$$

where C^{lc} is the gain reduction factor in NAMM to address the overlapping between two masking factors. In our letter, C^{lc} is set as 0.3, same as in [1].

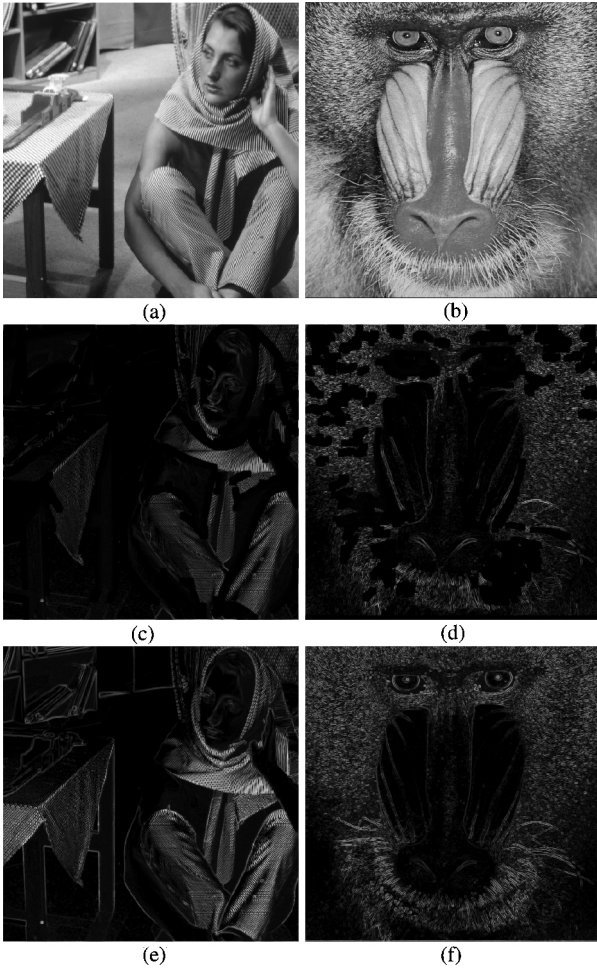


Fig. 5. CM in different JND models (higher brightness means a larger masking value). (a) Original *Barbara* image. (b) Original *Mandrill* image. (c) Model [1] with *Barbara*. (d) Model [1] with *Mandrill*. (e) Proposed model with *Barbara*. (f) Proposed model with *Mandrill*.

IV. OVERALL EXPERIMENTAL RESULTS

In this section, we will compare the proposed scheme's overall performance against the two latest direct pixel domain models (i.e., Yang *et al.*'s model [1] and Chou and Li's model [13]) with different images and application scenarios, as well as two different subjective viewing tests (one has been done in our laboratory while the other is publicly available).

A. Result of Noise Shaping

A better JND model should be able to guide shaping (i.e., hiding) more noise into an image at a certain level of resultant perceived images quality. The noise can be regulated according to the JND profile [1], [13]. At the same perceptual quality, the higher the injected-noise energy (measured by PSNR), the more accurate is the JND model. Ten images (with different visual content and spatial complexity) were used in this experiment, as listed in the first column of Table I.

For a comprehensive evaluation, the subjective viewing tests were conducted based on "adjectival categorical judgment methods" recommended by ITU-R BT.500-11 standard [16]. In each test, two images of the same scene (i.e., an image processed by the proposed model and that processed by one

TABLE I
SUBJECTIVE QUALITY EVALUATION RESULTS (THE PROPOSED MODEL AGAINST EACH OF THOSE IN [13] AND [1]) AND PSNRs FOR TEN IMAGES WITH DIFFERENT VISUAL CONTENT AND SPATIAL COMPLEXITY

Image	MOS (Against the Image Processed by the Proposed Model)				PSNR (dB) (as Indication of Perceptual Redundancy)		
	Mean		Standard Deviation		Proposed	Model [13]	Model [1]
	Model [13]	Model [1]	Model [13]	Model [1]			
<i>Airplane</i>	0.048	0.286	1.071	0.845	32.65	32.74	34.80
<i>Barbara</i>	0.143	0.333	1.062	0.856	29.93	31.64	31.35
<i>Boat</i>	0.429	-0.095	0.870	0.831	31.37	32.55	32.62
<i>Couple</i>	0.190	-0.048	0.750	0.590	32.43	33.57	32.71
<i>Gold</i>	-0.619	0.000	0.669	1.049	31.16	32.32	31.97
<i>Lena</i>	-0.286	-0.048	0.784	0.669	31.89	32.80	32.72
<i>Mandrill</i>	-0.190	-0.333	0.602	0.856	29.77	32.99	32.54
<i>Peppers</i>	-0.190	0.000	0.512	0.837	30.12	30.78	30.79
<i>Splash</i>	0.190	-0.143	0.928	1.014	30.96	31.43	31.35
<i>Tank</i>	0.143	-0.048	0.854	1.117	34.22	35.22	34.88
Average	-0.014	-0.010	0.810	0.866	31.45	32.60	32.58
Average additional redundancy (in dB) of our model						1.15	1.13

TABLE II
SCORES FOR SUBJECTIVE QUALITY EVALUATION

Subjective score	0	1	2	3
Description	Same quality	Only slightly better	Better	Much better

of the other models) were juxtaposed on the screen randomly at left or right. Twenty-one subjects were asked which image (left or right) is better and how much better it is by using quantitative scores listed in Table II, for all the image pairs.

The mean and standard deviation of the subjective scores of each model under comparison (i.e., Chou and Li's model [13], and Yang *et al.*'s model [1]) against the proposed one are given in the second to fifth columns of Table I. A negative (or positive) subjective score indicates that the image with the proposed model has better (or worse) perceptual quality than that with the model under comparison (either [13] or [1]), and its magnitude represents the extent of quality improvement (degradation). From Table I, we can see that the overall subjective quality tends to be of a near-zero negative mean of -0.014 and -0.010 , respectively. That is, the subjective quality for the images noised with the proposed JND model is very close (in fact, being slightly better) when compared to that with other models.

The PSNR of noised images is then used to measure the amount of noise being added by the JND models under comparison. The sixth to eighth columns of Table I show the PSNRs with the three JND profiles. The proposed model is better than existing relevant JND models, with the evidence of an average additional PSNR redundancy of 1.15 dB and 1.13 dB from the model in [13] and the one in [1], respectively, without jeopardy of visual quality (in conjunction with the subject results that we have just discussed). Since the images used in Table I are with both textual and not-so-textural visual content, the results confirm that the proposed scheme works better for both situations over the existing schemes.

B. Application in Visual Quality Gauging

To further demonstrate the proposed model, we apply it to a JND-based visual quality gauging metric that we developed earlier [4]. The metric is denoted as D , and is designed for

TABLE III
PREDICTION PERFORMANCE FOR DIFFERENT APPROACHES

Metric	R_P (with CI)	R_s (with CI)	R_e
$D-C$	0.8950 ([0.8091, 0.9435])	0.8161 ([0.6766, 0.8991])	17.13
$D-Y$	0.8885 ([0.7978, 0.9399])	0.8238 ([0.6891, 0.9035])	18.97
$D-P$	0.9028 ([0.8226, 0.9478])	0.8173 ([0.6785, 0.8997])	13.21

image and video with the provision of existence of enhanced edges. The image quality database in [17] is adopted (as in [4]), which gives subjective distortion scores with ten images and four processed (i.e., demosaicked) versions (inclusive of enhanced) for each image. The metric D in [4] evaluates and distinguishes noticeable contrast changes for visual quality assessment, using a JND model. Since the performance of D depends upon the accuracy of JND estimation, a better JND model will lead to better image quality gauging in line with subjective viewing scores given in [17]. Denote $D-P$, $D-Y$, and $D-C$ as the metric D with the proposed JND model, Yang *et al.*'s JND model, and Chou and Li's JND model, respectively. The comparison is in Table III, where the fit of goodness with the subjective scores [17] is quantitatively calculated using Pearson linear correlation coefficient (R_P , for prediction accuracy), Spearman rank order correlation coefficient (R_s , for monotonicity), and root mean squared error (R_e) between the subjective distortion scores and the predictive ones. For a perfect match between the objective and subjective scores, $R_P = R_s = 1$ and $R_e = 0$. A better quality metric has higher R_P and R_s , and lower R_e .

From Table III [with 95% confidence interval (CI) indicated for R_P and R_s], it has been found that the proposed JND model-based metric performs best among all the approaches in terms of R_P (prediction accuracy) and R_e , and R_s (monotonicity) of the proposed model-based metric is better than Chou and Li's one and only slightly lower than that of the best one (i.e., Yang *et al.*'s model). Therefore, we demonstrate again that the proposed JND model yields better overall performance than the existing relevant JND models.

V. CONCLUSION

In this letter, a new direct pixel domain JND estimator was proposed based upon the in-depth analysis on the existing relevant approaches. The TV-based image decomposition model was, therefore, used to improve the accuracy of CM evaluation (via decomposition into structural and textural images), as an attempt for overcoming the major shortcoming of the existing models. With this, we were able to differentiate CM effect due to edge and textured regions, in a well-grounded manner. We have, therefore, avoided underestimation of the visibility thresholds on texture, and provided a better and more comprehensive approach to mimic the masking effect of the HVS perception.

Proper subjective tests (both in-house and publicly accessible) have confirmed that the proposed scheme provided a more

accurate JND profile. At the same level of perceptual quality, the proposed model allowed more data redundancy (lower PSNR) in noise-shaping experiments when compared with the most related existing methods. We have also presented the application of the proposed JND estimator in visual distortion evaluation, and further demonstrated the improvement of the proposed model.

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